



NORTH SYDNEY GIRLS HIGH SCHOOL

HSC Mathematics Assessment Task 1

Term 4, 2011

Name: _____ Mathematics Class: _____

Time Allowed: 50 minutes + 2 minutes reading time

Total Marks: 50

Instructions:

- Attempt all five questions.
- Questions are of equal value.
- Start each question in a new booklet. Put your name on every booklet.
- Show all necessary working.
- Marks may be deducted for incomplete or poorly arranged work.
- Work down the page.
- Do not work in columns.
- Each question will be collected separately. Submit a blank booklet if you do not attempt a question.

Question	1ai	1aii	1aiii-b	2abc	2d	3	4	5a	5bc	Total
P4			/8	/7			/10	/4		/29
H6	/1				/3	/10				/14
H7		/1								/1
H9									/6	/6
										/50

Question 1 (10 marks)**Marks**

(a) Multiple choice. Write down the correct alternative on your writing paper.

(i) Part of the curve $y = f(x)$ is increasing and concave down.

1

Which of the following statements is true for this part of the curve?

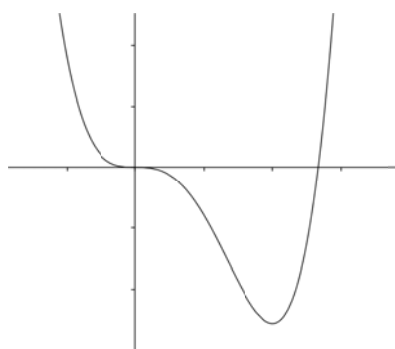
(A) $\frac{dy}{dx} > 0$ and $\frac{d^2y}{dx^2} > 0$

(C) $\frac{dy}{dx} < 0$ and $\frac{d^2y}{dx^2} < 0$

(B) $\frac{dy}{dx} < 0$ and $\frac{d^2y}{dx^2} > 0$

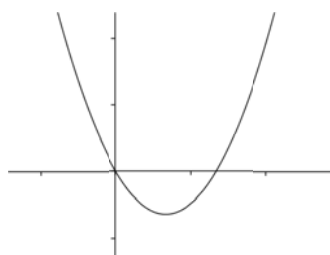
(D) $\frac{dy}{dx} > 0$ and $\frac{d^2y}{dx^2} < 0$

(ii) The graph $y = f(x)$ is shown below.

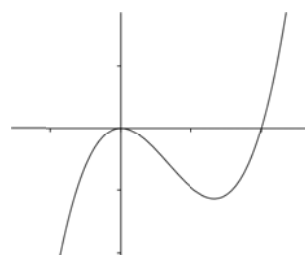
1

The graph of $y = f''(x)$ could be

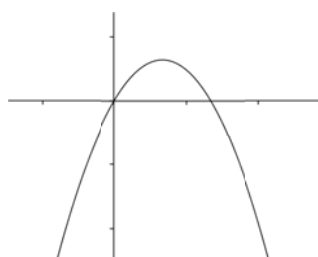
(A)



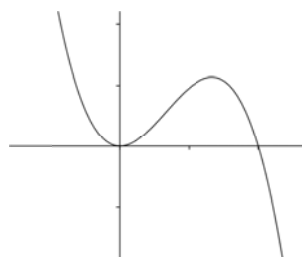
(C)



(B)



(D)



Question 1 continued**Marks**

- (iii) The quadratic equation $x^2 - 6x - 9 = 0$ has roots α and β . **1**

α and β are

- (A) real and equal.
- (B) real, unequal and rational.
- (C) real, unequal and irrational.
- (D) not real.

- (iv) The locus of the set of points whose distance from the x -axis is twice the distance from the y -axis has its equation given by **1**

- (A) $y = 2x$
- (B) $x = 2y$
- (C) $x^2 = 4y^2$
- (D) $y^2 = 4x^2$

- (v) The parabola $(x - 2)^2 = -8(y + 3)$ has

- (A) its vertex at $(2, -3)$ and directrix at $y = -1$
- (B) its vertex at $(2, -3)$ and directrix at $y = -5$
- (C) its focus at $(2, -3)$ and directrix at $y = -1$
- (D) its focus at $(2, -3)$ and directrix at $y = -5$

- (b) Show all necessary working.

If α and β are the roots of the equation $2x^2 - 5x - 10 = 0$, find the values of

- (i) $\alpha + \beta$ **1**
- (ii) $\alpha\beta$ **1**
- (iii) $(1 + \alpha)(1 + \beta)$ **1**
- (iv) $\alpha^2 + \beta^2$ **2**

Question 2 (10 marks)	Start a new booklet	Marks
(a) Form the quadratic equation with integer coefficients whose roots are $2 + \frac{\sqrt{3}}{2}$ and $2 - \frac{\sqrt{3}}{2}$.		2
(b) Find real numbers A , B and C , if $2x^2 + 5x - 19 \equiv A(x+3)(x-1) + B(x+3) + C(x-1)$.		3
(c) Show that the parabola $y = 2x^2 - 2kx + (3k^2 + 1)$ is positive definite.		2
(d) The function $f(x) = x^4 + ax^3 + bx + 1$ has a point of inflexion at $(2, 3)$. Find the values of a and b .		3

Question 3 (10 marks) Start a new booklet

Let $f(x) = 8x^2 - x^4$.

(a) Find the coordinates of the points where the curve crosses the axes.	2
(b) Find the coordinates of the stationary points and determine their nature.	4
(c) Find the coordinates of the points of inflexion.	2
(d) Sketch the graph of $y = f(x)$ indicating clearly the intercepts, stationary points and points of inflexion.	2

Question 4 (10 marks)

Start a new booklet

Marks

- (a) By making a suitable substitution or otherwise solve the equation

3

$$\left(x + \frac{1}{x}\right)^2 - 7\left(x + \frac{1}{x}\right) + 10 = 0.$$

Leave your answer(s) in exact form.

- (b) Find the values of k for which the equation $x^2 - 2kx + 4k - 3 = 0$ has two real and distinct roots.

3

- (c) Consider the parabola $y^2 = 2x - 4y - 2$.

(i) Find the coordinates of the vertex.

2

(ii) Find the coordinates of the focus.

1

(iii) Find the equation of the directrix.

1**Question 5** (10 marks)

Start a new booklet

- (a) The point $P(x, y)$ moves so that the sum of the squares of the distances from each of the points $A(0, 6)$ and $B(8, 0)$ is equal to 82.

4

Show that the locus of $P(x, y)$ is a circle and find its centre and radius.

- (b) Let $f(x) = (x - 2)^4 - 32x - 7$.

3

Does $y = f(x)$ have any points of inflexion? Justify your answer.

- (c) The roots of the equation $(p + q)x^2 - 6qx - 3p = 0$ are reciprocals.

3

Show that the sum of the roots is 8.

End of paper

MATHEMATICS TERM 4 2011 SOLUTIONS

Question 1.

(a)

(i) D

(ii) A

(iii) C

(iv) D

(v) A

(b)

$$2x^2 - 5x - 10 = 0$$

$$(i) \quad \alpha + \beta = -\frac{b}{a} = \frac{5}{2}$$

$$(ii) \quad \alpha\beta = \frac{c}{a} = \frac{-10}{2} = -5$$

$$\begin{aligned}(iii) \quad & (1+\alpha)(1+\beta) \\ &= 1 + (\alpha + \beta) + \alpha\beta \\ &= 1 + \frac{5}{2} - 5 \\ &= -\frac{3}{2}\end{aligned}$$

$$\begin{aligned}(iv) \quad & \alpha^2 + \beta^2 \\ &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= \left(\frac{5}{2}\right)^2 - 2 \times (-5) \\ &= \frac{25}{4} + 10 \\ &= \frac{65}{4}\end{aligned}$$

Question 2.

(a)

$$\alpha + \beta = 2 + \frac{\sqrt{3}}{2} + 2 - \frac{\sqrt{3}}{2} = 4$$

$$\alpha\beta = \left(2 + \frac{\sqrt{3}}{2}\right)\left(2 - \frac{\sqrt{3}}{2}\right) = 4 - \frac{3}{4} = \frac{13}{4}$$

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$x^2 - (4)x + \frac{13}{4} = 0$$

$$4x^2 - 16x + 13 = 0$$

(b)

$$2x^2 + 5x - 19 \equiv A(x+3)(x-1) + B(x+3) + C(x-1)$$

$$\text{coefficient } x^2: \quad A = 2$$

$$x = 1: -12 = 4B$$

$$B = -3$$

$$x = 0: -19 = -3A + 3B - C$$

$$-19 = -6 - 9 - C$$

$$C = 4$$

$$(c) \quad y = 2x^2 - 2kx + (3k^2 + 1)$$

$$\Delta = (2k)^2 - 4 \times 2 \times (3k^2 + 1)$$

$$= 4k^2 - 24k^2 - 8$$

$$= -(20k^2 + 8)$$

which is negative for all k .

also $a = 2 > 0$.

$\therefore$ positive definite

$$(d) \quad f(x) = x^4 + ax^3 + bx + 1$$

$$f'(x) = 4x^3 + 3ax^2 + b$$

$$f''(x) = 12x^2 + 6ax$$

$$f''(2) = 12 \times (2)^2 + 6a \times 2 = 0$$

$$\therefore 12a + 48 = 0$$

$$a = -4$$

$$f(2) = (2)^4 - 4(2)^3 + 2b + 1 = 3$$

$$16 - 32 + 2b + 1 = 3$$

$$2b = 18$$

$$b = 9$$

Question 3.

$$f(x) = 8x^2 - x^4$$

(a)

$$8x^2 - x^4 = 0$$

$$x^2(8 - x^2) = 0$$

$$\therefore x = 0, \pm\sqrt{8} \quad (= \pm 2\sqrt{2})$$

$$(0, 0) \quad (2\sqrt{2}, 0) \quad (-2\sqrt{2}, 0)$$

(b)

$$f'(x) = 16x - 4x^3 = 4x(4 - x^2)$$

$$f''(x) = 16 - 12x^2 = 4(4 - 3x^2)$$

$$\text{stat.pts when } f'(x) = 0$$

$$4x(4 - x^2) = 0$$

$$\therefore x = 0, 2, -2$$

$$\text{at } x = 0, y = 0, f''(0) = 16 > 0$$

$$\therefore \text{concave up. Minimum TP at } (0, 0)$$

$$\text{at } x = -2, y = 16, f''(-2) = -32 < 0$$

$$\therefore \text{concave down. Maximum TP at } (-2, 16)$$

$$\text{at } x = 2, y = 16, f''(2) = -32 < 0$$

$$\therefore \text{concave down. Maximum TP at } (2, 16)$$

(c)

$$\text{For inflexions } f''(x) = 0$$

$$16 - 12x^2 = 0$$

$$x = \pm\sqrt{\frac{16}{12}} = \pm\frac{2}{\sqrt{3}}$$

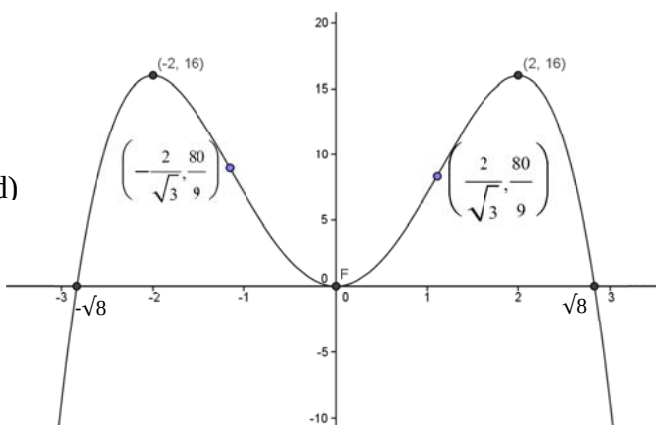
$$y = 8 \times \frac{4}{3} - \left(\frac{4}{3}\right)^2 = \frac{80}{9} = 8\frac{8}{9}$$

$$\therefore \left(\pm\frac{2}{\sqrt{3}}, 8\frac{8}{9}\right)$$

x	-2	$-2/\sqrt{3}$	0	$2/\sqrt{3}$	2
y''	-32	0	16	0	-32

$\therefore$ concavity changes-inflexions

(d)



Question 4.

(a)

$$\left(x + \frac{1}{x}\right)^2 - 7\left(x + \frac{1}{x}\right) + 10 = 0.$$

$$\text{Let } U = x + \frac{1}{x}$$

$$U^2 - 7U + 10 = 0$$

$$(U - 2)(U - 5) = 0$$

$$U = 2, 5$$

$$x + \frac{1}{x} = 2, \quad x + \frac{1}{x} = 5$$

$$x^2 + 1 = 2x, \quad x^2 + 1 = 5x$$

$$x^2 - 2x + 1 = 0, \quad x^2 - 5x + 1 = 0$$

$$(x - 1)^2 = 0, \quad x = \frac{5 \pm \sqrt{25 - 4 \times 1}}{2}$$

$$x = 1, \frac{5 \pm 21}{2}$$

(b)

$$x^2 - 2kx + 4k - 3 = 0$$

$$\Delta = (-2k)^2 - 4 \times 1 \times (4k - 3)$$

$$= 4k^2 - 16k + 12$$

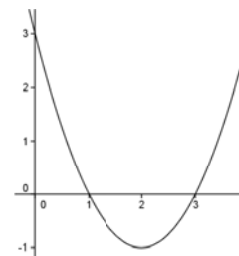
For real, distinct roots $\Delta > 0$

$$4k^2 - 16k + 12 > 0$$

$$k^2 - 4k + 3 > 0$$

$$(k - 1)(k - 3) > 0$$

$$\therefore k < 1 \text{ or } k > 3$$



(c)

$$y^2 = 2x - 4y - 2$$

(i)

$$y^2 + 4y = 2x - 2$$

$$y^2 + 4y + 4 = 2x + 2$$

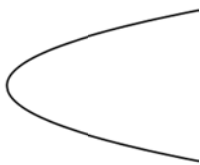
$$(y + 2)^2 = 2(x + 1)$$

$$\therefore \text{vertex } (-1, -2)$$

(ii)

Focal length = 0.5,

shape



$$\therefore \text{Focus } (-0.5, -2)$$

(iii)

$$\text{Directrix: } x = -1.5$$

Question 5.

(a)

Let $P(x, y)$ satisfy condition

$$PA^2 + PB^2 = 82$$

$$(x - 0)^2 + (y - 6)^2 + (x - 8)^2 + (y - 0)^2 = 82$$

$$x^2 + y^2 - 12y + 36 + x^2 - 16x + 64 + y^2 = 82$$

$$2x^2 - 16x + 2y^2 - 12y + 100 = 82$$

$$x^2 - 8x + y^2 - 6y = -9$$

$$x^2 - 8x + 16 + y^2 - 6y + 9 = -9 + 16 + 9$$

$$(x - 4)^2 + (y - 3)^2 = 4^2$$

Circle centre $(4, 3)$ radius 4

(b)

$$f(x) = (x - 2)^4 - 32x - 7$$

$$f'(x) = 4(x - 2)^3 - 32$$

$$f''(x) = 12(x - 2)^2$$

Possible inflexion at $f''(x) = 0$

$$\text{i.e. } x = 2$$

Check change of concavity

x	1	2	3
y''	12	0	12

No change in concavity. No inflexions.

(c)

$$(p + q)x^2 - 6qx - 3p = 0$$

let roots α, β

$$\alpha\beta = \frac{-3p}{p + q}$$

$$\text{but } \beta = \frac{1}{\alpha}$$

$$\therefore \frac{-3p}{p + q} = 1$$

$$-3p = p + q$$

$$q = -4p$$

$$\alpha + \beta = -\frac{-6q}{p + q}$$

$$= \frac{6 \times (-4p)}{p - 4p}$$

$$= \frac{-24p}{-3p}$$

$$= 8 \quad \text{as required}$$